Supplementary Material: Risk-Aware Multiobjective Evolutionary Optimization for Fast, Stable DeePC Tuning

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1 Purpose and organization of the supplementary material

This supplementary document supports the main paper in five ways. First, it clarifies why the proposed inner reduced-order DeePC formulation is more aligned with deployment-oriented outer optimization than a standard reduced-order template. Second, it gives the full rollout metrics, constraint definitions, signed-margin diagnostics, and algorithmic details that are only summarized in the main paper. Third, it reports additional system-wise ablation results so that the component-wise conclusions in the main paper are not supported by a single benchmark alone. Fourth, it provides representative tuned controller profiles for all four systems, which supports the interpretation that the proposed method solves a structural multiobjective design problem rather than a smooth scalar tuning problem. Fifth, it reports system-wise local-sensitivity summaries that support the main paper's claim that the search landscape contains threshold-driven regime changes.

For ease of reading, all equations, tables, and algorithms in this supplement use the prefix *S*. References such as "Eq. (5) in the main paper" or "Section 3.4 of the main paper" refer to the camera-ready main manuscript.

2 Why the proposed inner DeePC formulation differs from a standard reduced-order template

The inner DeePC stage problem in Eq. (5) of the main paper differs deliberately from a plain reduced-order DeePC template [5]. These differences are important because the outer optimization problem in the main paper is not nominal one-shot control tuning; it is a stochastic bi-objective design problem that explicitly targets adverse-tail control behavior and runtime.

First, the matrix $W_{tail}(\rho)$ introduces horizon-dependent tracking emphasis. A standard uniform stage penalty can produce a mismatch between the controller optimized inside the stage problem and the rollout-level design objective optimized outside. The tail-weighting matrix reduces this mismatch by biasing the inner controller toward sustained horizon-wide tracking rather than only transient early-horizon improvement.

Second, the slack penalty $\beta_h\|\psi\|_2^2$ encodes a preference for interior solutions with headroom. This matters because output safety in deployment is not only about avoiding hard violations. Designs that operate too close to the boundary are often fragile under perturbations even when they remain nominally feasible. The headroom term therefore improves the alignment between the inner controller and the signed-margin feasibility analysis used later in Eqs. (12)–(13).

Third, the move-suppression term $\beta_{\Delta u}\|E_1\bar{U}_f(\xi)\hat{g} - u_{k-1}\|_2^2$ reduces chattering and improves deployability. This term becomes especially important in the present tuning problem because rank transitions and stochastic perturbations can create candidate controllers with similar nominal tracking cost but substantially different actuation smoothness.

Fourth, the normalized regularization terms

$$\frac{\lambda_g}{r_a(\xi)}\|\hat{g}\|_2^2 + \frac{\lambda_\sigma}{pT_{ini}}\|\sigma\|_2^2$$

make regularization comparable across candidates with different reduced dimensions and different initialization horizons. Without these normalizations, the same nominal regularization parameters would induce systematically different effective penalties across controller regimes.

These four choices explain why the enriched controller in the main paper improves control quality, tail behavior, and front density in the factorial study. They also explain why the ablation results in Tables 2–4 degrade when key controller-side or evaluator-side ingredients are removed.

3 Extended rollout evaluation and risk-aware objective construction

Each candidate design ξ is evaluated through repeated stochastic closed-loop simulation. In scenario $s \in \{1, \ldots, S\}$, the reduced-order DeePC controller defined by Eq. (5) of the main paper is executed over a finite horizon T_{sim}, yielding trajectories

$$\{u_k^{(s)}(\xi), y_k^{(s)}(\xi), x_k^{(s)}(\xi)\}_{k=1}^{T_{sim}}$$

and per-step runtime traces

$$\tau_k^{(s)}(\xi) = t_{solve,k}^{(s)}(\xi) + t_{rebuild,k}^{(s)}(\xi).$$

The scenario-level rollout metrics are defined as follows.

The accumulated tracking error is

$$\mathrm{IAE}^{(s)}(\xi) = \sum_{k=1}^{T_{sim}} \epsilon_k^{(s)}(\xi), \tag{1}$$

where $\epsilon_k^{(s)}(\xi)$ is the normalized tracking error used in the implementation.

The accumulated input-energy surrogate is

$$U_2^{(s)}(\xi) = \sum_{k=1}^{T_{sim}} \|u_k^{(s)}(\xi)\|_2^2. \tag{2}$$

The accumulated input-variation term is

$$\delta U_2^{(s)}(\xi) = \sum_{k=1}^{T_{sim}} \|u_k^{(s)}(\xi) - u_{k-1}^{(s)}(\xi)\|_2^2. \tag{3}$$

Let $\Pi_{\mathcal{U}}(\cdot)$ denote projection onto the admissible input box. The accumulated input-saturation violation is

$$V_u^{(s)}(\xi) = \sum_{k=1}^{T_{sim}} \left\|u_k^{(s),*}(\xi) - \Pi_{\mathcal{U}}(u_k^{(s),*}(\xi))\right\|_1. \tag{4}$$

The accumulated output-safety violation is

$$V_y^{(s)}(\xi) = \sum_{k=1}^{T_{sim}} \left\|[|y_k^{(s)}(\xi)| - y_{\max}]_+\right\|_1. \tag{5}$$

Let $\tau_{tail} \subseteq \{1, \ldots, T_{sim}\}$ denote the final rollout fraction used to assess late-horizon behavior. The tail-tracking statistic is

$$e_{tail}^{(s)}(\xi) = \frac{1}{|\tau_{tail}|} \sum_{t \in \tau_{tail}} \epsilon_t^{(s)}(\xi). \tag{6}$$

These scenario-level quantities are combined into the control loss

$$L_{ctrl}^{(s)}(\xi) = \mathrm{IAE}^{(s)}(\xi) + \beta_u U_2^{(s)}(\xi) + \beta_\delta \delta U_2^{(s)}(\xi) + \beta_{V_u} V_u^{(s)}(\xi) + \beta_{V_y} V_y^{(s)}(\xi) + \beta_{tail} e_{tail}^{(s)}(\xi). \tag{7}$$

The risk-aware control objective in the main paper is then

$$J_{ctrl}^{risk}(\xi) = \mathrm{mean}_s\left(L_{ctrl}^{(s)}(\xi)\right) + \alpha_{ctrl}\,\mathrm{CVaR}_q\left(\{L_{ctrl}^{(s)}(\xi)\}_{s=1}^S\right) + \eta_h \phi(h(\xi)), \tag{8}$$

and the risk-aware runtime objective is

$$J_{time}^{risk}(\xi) = \mathrm{median}_{s,k}\left(\tau_k^{(s)}(\xi)\right) + \alpha_{time}\,\mathrm{CVaR}_q\left(\{\tau_k^{(s)}(\xi)\}_{s,k}\right). \tag{9}$$

Algorithm 1 implements this complete candidate-evaluation pipeline. This explicit rollout construction is what allows the main paper to make deployment-facing claims about TailC, TailT, and adverse-tail robustness rather than only about nominal controller cost.

4 Constraint model, signed margins, and feasibility diagnostics

The outer search is constrained by numerical-conditioning, safety, and failure-related requirements. Let $\kappa_{max}(\xi)$, $\sigma_{r,\min}(\xi)$, $fail_{max}(\xi)$, $x_{peak}(\xi)$, $y_{peak}(\xi)$, and $e_{tail}(\xi)$ denote the worst-case observed condition number, minimum retained singular value, rollout failure rate, peak state excursion, peak output excursion, and tail-tracking statistic across scenarios, respectively. The normalized constraint functions are

$$\begin{aligned}
c_\kappa(\xi) &= \log_{10}\left(\frac{\kappa_{max}(\xi)}{\kappa_{lim}}\right), & c_\sigma(\xi) &= \frac{\sigma_{lim} - \sigma_{r,\min}(\xi)}{\sigma_{lim}}, \\
c_{fail}(\xi) &= \frac{fail_{max}(\xi) - fail_{lim}}{fail_{lim}}, & c_x(\xi) &= \frac{x_{peak}(\xi) - x_{\max}}{x_{\max}}, \\
c_y(\xi) &= \frac{y_{peak}(\xi) - y_{\max}}{y_{\max}}, & c_{tail}(\xi) &= \frac{e_{tail}(\xi) - e_{tail,\max}}{e_{tail,\max}}.
\end{aligned} \tag{10}$$

The aggregate violation used by the evolutionary solvers is

$$CV(\xi) = \sum_{j \in \{\kappa,\sigma,fail,x,y,tail\}} \max\{0, c_j(\xi)\}. \tag{11}$$

A candidate is feasible if and only if $CV(\xi) = 0$.

For analysis, we also use signed headroom margins whose boundary is always at zero:

$$\begin{aligned}
d_\kappa(\xi) &= -c_\kappa(\xi), & d_\sigma(\xi) &= 1 - \frac{\sigma_{lim}}{\sigma_{r,\min}(\xi)}, \\
d_{fail}(\xi) &= 1 - \frac{fail_{max}(\xi)}{fail_{lim}}, & d_x(\xi) &= 1 - \frac{x_{peak}(\xi)}{x_{\max}}, \\
d_y(\xi) &= 1 - \frac{y_{peak}(\xi)}{y_{\max}}, & d_{tail}(\xi) &= 1 - \frac{e_{tail}(\xi)}{e_{tail,\max}}.
\end{aligned} \tag{12}$$

We then summarize the minimum safety headroom as

$$h(\xi) = \min\{d_\kappa(\xi), d_\sigma(\xi), d_{fail}(\xi), d_x(\xi), d_y(\xi), d_{tail}(\xi)\}. \tag{13}$$

Positive values indicate interior feasible margin, zero indicates boundary contact, and negative values indicate violation. These diagnostics support the headroom-shaping term in the main paper and are also used in the regime descriptor and sensitivity summaries reported later in Tables 9–12.

5 Extended evolutionary operator mechanics

The present search problem is not a standard continuous black-box benchmark. The design vector ξ mixes structural integers (T_{ini}, N, Δ) with continuous variables $(\theta_g, \theta_\sigma, \theta_{thr})$, the evaluation map is regime-switching because the retained reduced order $r_a(\xi)$ changes through singular-value thresholding, each evaluation is expensive and noisy because it requires repeated stochastic rollouts, and the desired output is a feasible Pareto set rather than one scalar optimum. These properties motivate constrained multiobjective evolutionary search [1,6,2,3,4], but they also require problem-aware adaptation of the operators.

5.1 Customized MOEA/D

We use MOEA/D because the problem is explicitly bi-objective and the main goal is to recover the control-computation trade-off surface. Let the weight vector of subproblem i be $\lambda^{(i)} = (\lambda_1^{(i)}, \lambda_2^{(i)})$ with $\lambda_1^{(i)}, \lambda_2^{(i)} \geq 0$ and $\lambda_1^{(i)} + \lambda_2^{(i)} = 1$. The distance between subproblems i and j is

$$d_{ij} = \|\lambda^{(i)} - \lambda^{(j)}\|_2, \tag{14}$$

and the T_n-nearest weight-space neighborhood is

$$B(i) = \{j \in \{1, \ldots, P\} : d_{ij} \text{ is among the } T_n \text{ smallest values}\}. \tag{15}$$

To avoid numerical imbalance between control loss and runtime, we use the normalized Tchebycheff scalarization

$$g^{(i)}(\xi) = \max_{l \in \{1,2\}} \lambda_l^{(i)} \left| \frac{\mathcal{F}_l(\xi) - z_l^{\star,(g)}}{s_l} \right|, \tag{16}$$

where $\mathcal{F}_1(\xi) = J_{ctrl}^{risk}(\xi)$, $\mathcal{F}_2(\xi) = J_{time}^{risk}(\xi)$, $z^{\star,(g)}$ is the current ideal point, and $s_l > 0$ is the objective scaling factor.

Plain differential mutation is not suitable for the mixed DeePC design vector. For three parent vectors $\xi_a^{(g)}$, $\xi_b^{(g)}$, and $\xi_c^{(g)}$, we therefore update the continuous part by

$$v^{(c)} = \xi_a^{(g,c)} + F_{DE}\big(\xi_b^{(g,c)} - \xi_c^{(g,c)}\big), \tag{17}$$

and the structural part by

$$v^{(d)} = \Pi_{\mathbb{Z}}\Big(\xi_a^{(g,d)} + F_{DE}\big(\xi_b^{(g,d)} - \xi_c^{(g,d)}\big)\Big), \tag{18}$$

followed by crossover and projection onto Ξ. This preserves directional search information while ensuring that offspring remain physically valid controller designs.

Replacement must also be feasibility-aware. Let $\tilde{\xi}$ denote an offspring and ξ an incumbent. The feasibility-first replacement relation is

$$\xi \prec_f \tilde{\xi} \iff \begin{cases} CV(\tilde{\xi}) = 0 \text{ and } CV(\xi) > 0, \quad \text{or} \\ CV(\tilde{\xi}) > 0,\ CV(\xi) > 0, \text{ and } CV(\tilde{\xi}) < CV(\xi), \quad \text{or} \\ CV(\tilde{\xi}) = CV(\xi) = 0 \text{ and } g^{(i)}(\tilde{\xi}) \leq g^{(i)}(\xi). \end{cases} \tag{19}$$

This modification is essential because the search space contains numerically attractive but infeasible designs near rank transitions and conditioning boundaries. Restricting replacement to weight-space neighborhoods also helps stabilize distinct combinations of reduced order, retained data length, and headroom rather than collapsing the search prematurely to one brittle regime.

5.2 Customized NSGA-II

We use NSGA-II because dominance-based search complements MOEA/D on nonconvex, irregular, or partially disconnected trade-off sets. However, NSGA-II also requires problem-aware adaptation.

The first change is feasibility-aware constrained dominance. For two candidates ξ and ζ,

$$\xi \prec_c \zeta \iff \begin{cases} CV(\xi) = 0 \text{ and } CV(\zeta) > 0, & \text{or} \\ CV(\xi) > 0,\ CV(\zeta) > 0, \text{ and } CV(\xi) < CV(\zeta), & \text{or} \\ CV(\xi) = CV(\zeta) = 0 \text{ and } \xi \prec \zeta, \end{cases} \tag{20}$$

where $\prec$ is ordinary Pareto domination on $\mathcal{F}(\xi)$. Constrained nondominated sorting with (20) produces fronts F_t.

The second change is regime-diversity crowding. Let the normalized regime descriptor be

$$\Gamma(\xi) = \left[\tilde{r}_a(\xi), \tilde{\Delta}(\xi), \tilde{h}(\xi)\right], \tag{21}$$

where each component is scaled to a comparable range within the current front. The regime-separation distance inside front F_t is

$$RD(\xi_i) = \min_{\xi_j \in F_t,\ j \neq i} \|\Gamma(\xi_i) - \Gamma(\xi_j)\|_1. \tag{22}$$

We then define the augmented crowding score

$$CD(\xi_i) = \sum_{l=1}^{2} \frac{\mathcal{F}_l\big(\xi_{i+1}^{(l)}\big) - \mathcal{F}_l\big(\xi_{i-1}^{(l)}\big)}{\mathcal{F}_l^{\max}(F_t) - \mathcal{F}_l^{\min}(F_t)} + \eta_{reg} RD(\xi_i). \tag{23}$$

This favors isolated feasible controller regimes that standard spatial crowding can under-preserve when dense low-rank regions dominate the local population geometry.

Tournament selection therefore uses constrained rank first and augmented crowding second:

$$\mathrm{Tour}(\xi_a, \xi_b) = \begin{cases} \xi_a, & \text{if } \xi_a \prec_c \xi_b, \\ \xi_b, & \text{if } \xi_b \prec_c \xi_a, \\ \xi_a, & \text{if } rank(\xi_a) = rank(\xi_b) \text{ and } CD(\xi_a) > CD(\xi_b), \\ \xi_b, & \text{if } rank(\xi_a) = rank(\xi_b) \text{ and } CD(\xi_b) > CD(\xi_a). \end{cases} \tag{24}$$

For structural variables, offspring components are produced by discrete recombination:

$$\tilde{\xi}_j^{(d)} = \begin{cases} Pa_j^{(1,d)}, & \text{with probability } 1/2, \\ Pa_j^{(2,d)}, & \text{with probability } 1/2, \end{cases} \qquad j \in \{1,2,3\}. \tag{25}$$

For continuous variables, we use SBX-style crossover:

$$\begin{aligned} \tilde{\xi}_{1,j}^{(c)} &= \frac{1}{2}\Big[(1+\gamma_j)Pa_j^{(1,c)} + (1-\gamma_j)Pa_j^{(2,c)}\Big], \\ \tilde{\xi}_{2,j}^{(c)} &= \frac{1}{2}\Big[(1-\gamma_j)Pa_j^{(1,c)} + (1+\gamma_j)Pa_j^{(2,c)}\Big], \end{aligned} \tag{26}$$

with

$$\gamma_j = \begin{cases} (2u_j)^{\frac{1}{\eta_c+1}}, & u_j \leq \frac{1}{2}, \\ \left(\frac{1}{2(1-u_j)}\right)^{\frac{1}{\eta_c+1}}, & u_j > \frac{1}{2}. \end{cases} \tag{27}$$

To cross regime boundaries, mutation on structural variables must occasionally be global rather than purely local. We therefore use

$$\tilde{\xi}_j^{(d)} \leftarrow \begin{cases} \mathrm{UnifInt}(\ell_j, u_j), & \text{with probability } p_{reset}, \\ \Pi_{\mathbb{Z}}(\tilde{\xi}_j^{(d)} + \varepsilon_j), & \text{with probability } 1 - p_{reset}, \end{cases} \tag{28}$$

where $\varepsilon_j \sim \mathcal{D}_j$ is a bounded local mutation. Continuous variables undergo polynomial mutation with probability p_m:

$$\tilde{\xi}_j^{(c)} = \tilde{\xi}_j^{(c)} + \Phi_j^{poly}(u_j - \ell_j). \tag{29}$$

Together, these changes make both MOEA/D and NSGA-II consistent with the mixed, feasibility-constrained, and regime-switching geometry of DeePC hyperparameter tuning. Algorithm 2 summarizes the full outer optimization pipeline.

6 Benchmark details

Table 1 lists the four benchmark systems used throughout the main paper and this supplement. The systems differ in input dimension, output dimension, admissible control range, and disturbance/noise scale. This heterogeneity is useful because it shows that the proposed framework is not tied to one specific physical configuration.

Table 1. Benchmark systems used in the experimental study.

System	n	m	p	Input limits	Noise bounds
S1 Vehicle Rollover	4	1	1	$[-90, 90]$	$d_p, d_m \leq 10^{-3}$
S2 LTV	4	2	2	$[-1, 1]$	$d_p, d_m \leq 2 \times 10^{-3}$
S3 Quadruple Tank	4	2	2	$[0, 10]$	$d_p, d_m \leq 10^{-3}$
S4 LFC	3	1	1	$[-0.2, 0.2]$	$d_p, d_m \leq 5 \times 10^{-5}$

7 Extended experimental protocol and how the supplementary tables support the main paper

7.1 Setup and evaluation protocol

For each system, each candidate controller design vector ξ is evaluated by stochastic closed-loop rollouts over T_{sim}. Algorithm 1 computes the scenario-level metrics in Eqs. (1)–(6), forms the control loss in Eq. (7), aggregates the rollout statistics into the risk-aware objectives in Eqs. (8) and (9), and evaluates feasibility through Eqs. (10)–(13). Thus, the reported values of J^{risk}_{ctrl}, J^{risk}_{time}, TailC, and TailT are rollout-based controller metrics, while HV and ndCount are search-quality indicators derived from the final feasible nondominated set.

All methods use the same rollout evaluator, the same stochastic scenario count, the same search ranges, and the same total evaluation budget. The default budget is $P(G+1) = 50 \times (20+1) = 1100$ candidate evaluations per run, and the reported suite results use 5 independent repeats unless otherwise stated. The robust outer evaluation uses $S = 10$ scenarios, a tail fraction of 0.2, and the resulting CVaR confidence level q. The search ranges are $T_{ini} \in [5, 50]$, $N \in [5, 100]$, $\Delta \in [100, 900]$, $\theta_g \in [-8, 4]$, $\theta_\sigma \in [-8, 8]$, and $\theta_{thr} \in [-10, -1]$, which are tightened using empirical quantiles with 10% padding after preliminary screening.

MOEA/D uses neighborhood size $T_n = 20$, differential factor $F_{DE} = 0.5$, and crossover rate $CR = 0.9$. NSGA-II uses crossover probability $p_c = 0.9$, mutation probability $p_m = 0.3$, SBX index $\eta_c = 15$, polynomial mutation index $\eta_m = 20$, structural reset probability $p_{reset} = 0.05$, and regime-diversity weight $\eta_{reg} = 0.1$. The inner controller uses $Q = 1000$, $R = 10^{-3}$, $\beta_{\Delta u} = 10^{-3}$, and $\beta_h = 10$. The rollout loss uses $\beta_u = 10^{-3}$, $\beta_\delta = 10^{-3}$, $\beta_{V_u} = 50$, $\beta_{V_y} = 1000$, and $\beta_{tail} = 1$, while the outer objective uses $\alpha_{ctrl} = 1.0$, $\eta_h = 1$, and $\alpha_{time} = 1.0$.

The random optimizer uses the same search box, rollout evaluator, stochastic scenarios, and total evaluation budget as MOEA/D and NSGA-II but samples candidates uniformly. The fixed design does not search the design space and is included only as a deployment-style reference point. These choices make the numerical comparisons in the main paper directly interpretable as optimizer-side rather than evaluator-side differences.

7.2 How the supplementary tables support the main findings

The supplementary tables are designed to strengthen three central claims of the main paper.

First, the component-wise importance of the full framework is not limited to the single ablation table shown in the main paper. Tables 2, 3, and 4 extend the ablation evidence to the remaining three systems. Together with the main-paper ablation table, they show that structural co-design, CVaR-based risk shaping, reduced-order adaptation, and appropriate time-side aggregation all contribute materially to performance.

Second, the structural interpretation of the final Pareto sets is not anecdotal. Tables 5–8 list representative control-oriented and runtime-oriented tuned designs for all four systems and all controller families. These tables show that high-control and low-runtime solutions differ systematically in horizons, retained data length, and truncation level, which supports the main paper's interpretation of DeePC tuning as structural multiobjective design.

Third, the regime-switching landscape argument is supported numerically beyond the heatmap figure shown in the main paper. Tables 9–12 summarize the dominant perturbation directions for each system. They show that perturbing T_{ini}, N, or σ_{thr} can trigger substantial changes in $r_a(\xi)$ and measurable jumps in both control loss and runtime, which is exactly the mechanism that motivates regime-preserving evolutionary search.

8 Extended ablation studies

Tables 2–4 extend the ablation analysis from the main paper to systems S2–S4. Each ablation removes or modifies one design choice while keeping the same rollout evaluator, the same feasibility model from Eqs. (10)–(13), and the same evolutionary search budget. The *Full* row is the complete method and serves as the reference point.

8.1 S2 ablation: LTV system

Table 2 shows that on S2 the full method remains strongest under both customized MOEA/D and customized NSGA-II. Fixing structural variables causes the largest degradation in HV, runtime, and adverse-tail behavior, which strongly supports the main paper's claim that DeePC tuning is fundamentally structural. Removing CVaR improves nominal time-side values but weakens overall frontier quality and worsens TailC/TailT, showing that explicit risk shaping is doing real work rather than merely re-scaling the objectives. Removing reduced-order adaptation or online updating weakens both frontier quality and control-runtime balance. The TMean and Single variants are also informative: they can make selected nominal statistics look attractive, but they are weaker from the standpoint of deployment-robust Pareto recovery.

Table 2. S2 ablation of the customized MOEA/D and NSGA-II frameworks.

Method	HV	J_{ctrl}^{risk}	J_{time}^{risk}	$ndCount$	$tailC$	$tailT$
			Customized MOEA/D			
Full	**184854.12** ± **67.15**	**2.14** ± **0.05**	2.75 ± 0.30	**50.00** ± **0.00**	**0.21** ± **0.00**	0.32 ± **0.03**
FixHor	161407.64 ± 862.60	15.27 ± 1.62	12.40 ± 0.84	49.00 ± 1.00	0.74 ± 0.08	0.77 ± 0.10
NoCVaR	179978.59 ± 889.00	2.78 ± 0.42	**0.16** ± **0.01**	40.00 ± 1.00	0.28 ± 0.04	0.29 ± 0.09
NoUpd	179994.05 ± 278.60	6.59 ± 1.53	2.70 ± 0.17	40.00 ± 1.00	0.31 ± 0.07	0.35 ± 0.05
NoRO	178930.77 ± 744.80	6.96 ± 1.27	3.29 ± 0.49	43.00 ± 2.00	0.33 ± 0.06	0.38 ± 0.07
Single	183895.10 ± 242.60	6.10 ± 0.69	0.68 ± 0.08	49.00 ± 8.00	0.28 ± 0.07	**0.29** ± 0.06
TMean	178728.85 ± 824.90	6.83 ± 2.42	3.42 ± 0.33	39.00 ± 1.00	0.32 ± 0.13	0.38 ± 0.10
			Customized NSGA-II			
Full	**184867.67** ± **47.09**	**2.40** ± **0.09**	3.01 ± 0.13	30.00 ± 4.00	**0.24** ± 0.01	0.36 ± **0.02**
FixHor	161372.68 ± 1763.00	14.83 ± 1.84	12.02 ± 0.49	28.00 ± 2.00	0.73 ± 0.07	0.69 ± 0.05
NoCVaR	179522.43 ± 367.62	2.83 ± 0.29	**0.15** ± **0.03**	25.00 ± **1.00**	0.28 ± 0.03	0.41 ± 0.06
NoUpd	179462.17 ± 510.00	5.64 ± 0.74	3.03 ± 0.30	**31.00** ± 3.00	0.26 ± 0.01	0.37 ± 0.04
NoRO	178360.42 ± 1011.00	6.53 ± 0.12	3.65 ± 0.61	20.00 ± 3.00	0.30 ± **0.01**	0.35 ± 0.03
Single	183774.33 ± 210.90	6.00 ± 0.39	0.83 ± 0.10	30.00 ± **1.00**	0.30 ± 0.04	**0.31** ± 0.17
TMean	177223.73 ± 730.80	6.49 ± 1.27	4.29 ± 0.34	29.00 ± 4.00	0.31 ± 0.07	0.36 ± 0.07

8.2 S3 ablation: Quadruple Tank system

Table 3 shows the same qualitative story on S3. The Full method is again strongest overall. FixHor produces the most visible collapse in HV and tail behavior, which reinforces the main paper's conclusion that horizon and retained-data co-design are central rather than incidental. NoCVaR and Single can appear nominally attractive on the time side, but this comes at the expense of a weaker robust frontier. NoRO consistently worsens runtime-related behavior, supporting the importance of search-controlled reduced-order truncation.

8.3 S4 ablation: LFC system

Table 4 shows the same pattern on S4. The full proposal achieves the best balance between HV, risk-aware control cost, runtime, and tail metrics. Fixing structural variables again produces the strongest degradation, confirming that structural co-design is indispensable. NoCVaR and Single reduce nominal runtime statistics but weaken robust frontier recovery. NoRO degrades both runtime and tail behavior, again confirming that search-controlled reduced-order adaptation is a key part of the final method rather than a cosmetic efficiency trick.

9 Structural controller profiles across all systems

This section links the representative tuned designs to the main paper's claim that DeePC tuning is a structural multiobjective design problem. Tables 5–8 report representative control-oriented and time-oriented tuned configurations for every system and optimizer. The broad pattern is consistent: control-oriented solutions usually select longer horizons and richer retained data windows, while runtime-oriented solutions prefer shorter horizons and more aggressive truncation. The

Table 3. S3 ablation of the customized MOEA/D and NSGA-II frameworks.

Method	HV	J_{ctrl}^{risk}	J_{time}^{risk}	$ndCount$	$tailC$	$tailT$
Customized MOEA/D						
Full	**485122.96 ± 382.70**	**2.20 ± 0.09**	2.76 ± 0.06	**49.00** ± 3.00	**0.22 ± 0.01**	0.35 ± 0.03
FixHor	402452.15 ± 5333.00	15.16 ± 2.05	13.32 ± 0.31	20.00 ± **0.00**	0.73 ± 0.08	0.68 ± 0.10
NoCVaR	469707.82 ± 2471.74	2.99 ± 0.45	**0.20 ± 0.03**	19.00 ± 4.00	0.30 ± 0.05	0.40 ± 0.11
NoUpd	466842.56 ± 2121.00	6.54 ± 0.23	3.24 ± 0.33	19.00 ± 1.00	0.30 ± 0.60	**0.31** ± 0.06
NoRO	463713.71 ± 4117.00	7.34 ± 0.53	3.76 ± 0.70	20.00 ± 1.00	0.33 ± 0.07	0.36 ± 0.07
Single	481861.55 ± 523.40	6.47 ± 0.44	0.74 ± 0.10	20.00 ± 1.00	0.29 ± 0.01	0.35 ± **0.01**
TMean	467547.94 ± 6138.00	6.73 ± 1.08	3.14 ± 1.06	20.00 ± 5.00	0.30 ± 0.06	0.34 ± 0.07
Customized NSGA-II						
Full	**485011.19 ± 140.22**	**2.23 ± 0.16**	2.78 ± 0.12	**38.00** ± 1.00	0.27 ± 0.16	0.35 ± **0.01**
FixHor	408478.78 ± 1460.00	14.93 ± 1.96	13.02 ± 0.10	36.00 ± 2.00	0.74 ± 0.09	0.77 ± 0.07
NoCVaR	467235.52 ± 2640.46	2.77 ± 1.45	**0.20 ± 0.02**	34.00 ± **0.00**	0.28 ± **0.02**	0.36 ± 0.04
NoUpd	466715.43 ± 2106.00	6.23 ± 0.68	3.28 ± 0.35	30.00 ± **0.00**	0.28 ± 0.04	0.36 ± 0.05
NoRO	464386.34 ± 1072.00	6.82 ± 0.60	3.69 ± 0.18	35.00 ± 2.00	0.31 ± 0.02	0.37 ± 0.06
Single	481084.71 ± 663.60	5.82 ± 0.94	0.88 ± 0.41	36.00 ± 3.00	**0.22** ± 0.09	**0.31** ± 0.15
TMean	463887.97 ± 522.30	6.60 ± 0.36	3.76 ± 0.10	31.00 ± 2.00	0.31 ± 0.03	0.43 ± 0.08

specific values differ by system, but the structural trade-off itself is stable across systems.

9.1 S1 structural profiles

Table 5 supports the main paper's interpretation of the S1 Pareto set. For both MOEA/D and NSGA-II, the best-control and best-time designs differ in more than one scalar value; they differ in initialization horizon, prediction horizon, retained data length, and truncation level. This shows that the control-runtime trade-off is tied to structural controller geometry.

9.2 S2 structural profiles

Table 6 shows that the same structural interpretation holds on S2. The runtime-oriented solutions use markedly smaller prediction horizons and often smaller retained data windows than the corresponding control-oriented solutions. The proposed method therefore does not merely tune one regularization coefficient; it selects different controller geometries for different parts of the Pareto set.

9.3 S3 structural profiles

Table 7 shows that the same interpretation persists on S3. The best-control solutions typically favor richer data windows or longer horizons, whereas the best-time solutions more often compress the predictive geometry. This again supports the main paper's claim that the Pareto set is structurally organized.

Table 4. S4 ablation of the customized MOEA/D and NSGA-II frameworks.

Method	HV	J_{ctrl}^{risk}	J_{time}^{risk}	*ndCount*	*tailC*	*tailT*
			Customized MOEA/D			
Full	**334988.54 ± 224.93**	**2.17 ± 0.07**	2.76 ± 0.18	**46.00** ± 3.00	**0.21 ± 0.00**	0.33 ± **0.03**
FixHor	281929.90 ± 3097.80	15.22 ± 1.83	12.86 ± 0.58	40.00 ± **0.00**	0.74 ± 0.08	0.73 ± 0.10
NoCVaR	324843.21 ± 1680.37	2.89 ± 0.44	**0.18 ± 0.02**	45.00 ± 2.00	0.29 ± 0.04	0.34 ± 0.10
NoUpd	323418.31 ± 1199.80	6.57 ± 0.88	2.97 ± 0.25	40.00 ± 1.00	0.30 ± 0.34	0.33 ± 0.06
NoRO	321322.24 ± 2430.90	7.15 ± 0.90	3.53 ± 0.60	**46.00** ± 1.00	0.33 ± 0.06	0.37 ± 0.07
Single	332878.33 ± 383.00	6.29 ± 0.57	0.71 ± 0.09	40.00 ± 4.00	0.29 ± 0.04	**0.32** ± 0.03
TMean	323138.40 ± 3481.45	6.78 ± 1.75	3.28 ± 0.69	39.00 ± 3.00	0.31 ± 0.09	0.36 ± 0.09
			Customized NSGA-II			
Full	**334939.43 ± 93.66**	**2.31 ± 0.13**	2.90 ± 0.13	35.00 ± 2.00	**0.25** ± 0.09	0.36 ± **0.02**
FixHor	284925.73 ± 1611.50	14.88 ± 1.90	12.52 ± 0.29	**37.00** ± 2.00	0.73 ± 0.08	0.73 ± 0.06
NoCVaR	323378.98 ± 1504.04	2.80 ± 0.87	**0.17 ± 0.02**	32.00 ± **1.00**	0.28 ± 0.02	0.39 ± 0.05
NoUpd	323088.80 ± 1308.00	5.93 ± 0.71	3.15 ± 0.32	35.00 ± **1.00**	0.27 ± 0.02	0.36 ± 0.04
NoRO	321373.38 ± 1041.50	6.68 ± 0.36	3.67 ± 0.39	32.00 ± 2.00	0.31 ± **0.01**	0.36 ± 0.05
Single	332429.52 ± 437.25	5.91 ± 0.67	0.86 ± 0.25	34.00 ± 2.00	0.26 ± 0.07	**0.31** ± 0.16
TMean	320555.85 ± 626.55	6.54 ± 0.82	4.02 ± 0.22	33.00 ± 3.00	0.31 ± 0.05	0.40 ± 0.08

9.4 S4 structural profiles

Table 8 confirms the same pattern on S4. Even when some controller families use similar horizons across both objectives, other variables such as retained data length, truncation threshold, or regularization still shift in a structured manner. This supports the claim that practical DeePC tuning is not a one-dimensional regularization problem.

10 Local sensitivity and landscape discontinuity across all systems

This section connects the numerical sensitivity summaries to the main paper's regime-switching argument. Tables 9–12 report the dominant perturbation directions for control loss, runtime, reduced order, and minimum safety headroom. Across systems, two patterns recur. First, perturbations of T_{ini}, N, and σ_{thr} frequently trigger nonzero $|\delta r_a|$, which confirms that the landscape contains threshold-driven dimensional transitions. Second, these regime transitions can be accompanied by large changes in both J_{ctrl}^{risk} and J_{time}^{risk}, which explains why continuous-only local tuning is structurally blind to the actual search geometry.

10.1 S1 sensitivity summary

Table 9 shows that on S1, both MOEA/D and NSGA-II identify T_{ini} and σ_{thr} as dominant perturbation directions, with nontrivial changes in both runtime and reduced order. This directly supports the main paper's Figure 3 interpretation for S1.

Algorithm 1: Evaluation of a risk-aware candidate controller

Input: Candidate ξ; past input-output data (u_{ini}, y_{ini}); reference trajectory r; pre-collected dataset (u^d, y^d); rollout horizon T_{sim}; number of scenarios S.

Output: Objective vector $\mathcal{F}(\xi)$; total constraint violation $CV(\xi)$; signed-margin diagnostics.

Initialize the scenario index set $\mathcal{S} = \{1, \ldots, S\}$.

foreach $s \in \mathcal{S}$ **do**

- Construct the Hankel matrix $H(\xi)$ from the data window of length T_{data}.
- Determine the retained reduced order $r_a(\xi)$ and form the predictive matrix $\hat{M}(\xi)$.
- Run a closed-loop rollout over T_{sim} steps using the inner controller from Eq. (5) of the main paper.
- Record the scenario metrics $\mathrm{IAE}^{(s)}$, $U_2^{(s)}$, $\delta U_2^{(s)}$, $V_u^{(s)}$, $V_y^{(s)}$, $e_{tail}^{(s)}$, and the runtime traces $\{\tau_k^{(s)}\}_{k=1}^{T_{sim}}$ using Eqs. (1)–(6).

end foreach

Compute the scenario-level losses $L_{ctrl}^{(s)}$ using Eq. (7).

Aggregate scenario losses into $J_{ctrl}^{risk}(\xi)$ and $J_{time}^{risk}(\xi)$ using Eqs. (8) and (9).

Compute the constraint values using Eq. (10), the aggregate violation using Eq. (11), and the signed margins using Eqs. (12)–(13).

return $\mathcal{F}(\xi)$, $CV(\xi)$, *and the signed-margin diagnostics.*

10.2 S2 sensitivity summary

Table 10 shows an even clearer role for T_{ini} and N on S2, with rank changes up to 4. This means that structural perturbations can induce large controller-regime shifts, which is exactly the kind of landscape behavior that motivates the proposed mixed-variable regime-preserving operators.

10.3 S3 sensitivity summary

Table 11 shows that on S3, T_{ini} and N remain important structural variables, while σ_{thr} still controls rank changes on the time side. The large perturbation effects again support the main paper's argument that the landscape is non-smooth and regime-switching.

10.4 S4 sensitivity summary

Table 12 shows that on S4, T_{ini} and N again dominate the structural side, while λ_σ and λ_g also influence the trade-off in a system-dependent way. Crucially, nonzero $|\delta r_a|$ still appears under small structural perturbations, reinforcing the regime-transition narrative.

Table 5. Representative control-oriented and runtime-oriented tuned configurations for S1.

Controller	Optimizer	$\mathcal{F}(\xi)$	$\mathbf{T_{ini}}$	$\mathbf{N}$	$\mathbf{\Delta}$	$\theta_{\mathbf{g}}$	θ_{σ}	$\theta_{\mathbf{thr}}$
C1	MOEA/D	J^{risk}_{ctrl}	30	20	324	-1.47	1.60	-4.28
		J^{risk}_{time}	35	32	241	-1.47	2.10	-3.40
C2		J^{risk}_{ctrl}	29	30	358	-0.87	1.00	-4.31
		J^{risk}_{time}	29	26	220	-2.17	0.70	-4.25
C3		J^{risk}_{ctrl}	30	19	359	-2.51	3.11	-6.53
		J^{risk}_{time}	22	10	291	-2.20	-1.39	-7.96
C4		J^{risk}_{ctrl}	22	12	481	0.60	2.00	-3.95
		J^{risk}_{time}	35	21	200	-1.26	3.70	-5.60
Ours		J^{risk}_{ctrl}	37	35	512	-0.60	-1.10	-1.60
		J^{risk}_{time}	32	10	349	-2.30	-1.20	-1.41
C1	NSGA-II	J^{risk}_{ctrl}	31	12	215	0.68	1.83	-3.26
		J^{risk}_{time}	23	25	307	0.57	1.53	-3.67
C2		J^{risk}_{ctrl}	30	16	218	-0.67	1.27	-2.73
		J^{risk}_{time}	22	11	215	-1.88	0.13	-3.06
C3		J^{risk}_{ctrl}	29	23	387	-1.94	-0.86	-5.58
		J^{risk}_{time}	24	12	551	0.01	-0.31	-6.86
C4		J^{risk}_{ctrl}	30	11	209	-0.85	1.45	-4.13
		J^{risk}_{time}	28	12	230	-0.75	1.78	-4.28
Ours		J^{risk}_{ctrl}	30	15	335	-0.18	1.70	-3.85
		J^{risk}_{time}	21	11	282	0.58	-5.24	-4.54
Fixed	N/A	J^{risk}_{ctrl} J^{risk}_{time}	35	45	200	-2.00	-2.00	-6.00

Table 6. Representative control-oriented and runtime-oriented tuned configurations for S2.

Controller	**Optimizer**	$\mathcal{F}(\xi)$	$\mathbf{T_{ini}}$	**N**	$\boldsymbol{\Delta}$	$\theta_{\mathbf{g}}$	θ_{σ}	$\theta_{\mathbf{thr}}$
C1	MOEA/D	J_{ctrl}^{risk}	10	23	212	-1.20	-3.68	-2.88
		J_{time}^{risk}	13	22	208	-1.34	0.60	-3.39
C2		J_{ctrl}^{risk}	10	21	200	-2.12	-2.24	-2.08
		J_{time}^{risk}	10	21	200	-2.31	-2.24	-2.08
C3		J_{ctrl}^{risk}	11	10	218	-2.55	-2.34	-6.69
		J_{time}^{risk}	11	10	218	-2.40	-2.31	-6.73
C4		J_{ctrl}^{risk}	11	42	215	-2.43	-1.96	-4.06
		J_{time}^{risk}	11	10	215	-2.43	-1.68	-4.27
Ours		J_{ctrl}^{risk}	10	14	314	-0.57	-2.65	-4.55
		J_{time}^{risk}	10	14	314	-0.38	-2.65	-4.55
C1	NSGA-II	J_{ctrl}^{risk}	15	37	349	-0.77	-6.00	-2.81
		J_{time}^{risk}	15	12	212	0.56	1.36	-3.34
C2		J_{ctrl}^{risk}	13	14	336	0.02	-5.71	-4.81
		J_{time}^{risk}	11	12	210	-0.98	0.57	-2.86
C3		J_{ctrl}^{risk}	14	18	338	-0.22	-5.04	-3.23
		J_{time}^{risk}	12	11	213	-0.42	-4.63	-5.47
C4		J_{ctrl}^{risk}	10	19	309	-0.04	-3.84	-5.22
		J_{time}^{risk}	13	15	205	-1.93	-5.25	-4.42
Ours		J_{ctrl}^{risk}	10	22	343	0.49	-5.35	-4.55
		J_{time}^{risk}	10	13	213	-1.09	-1.70	-2.00
Fixed	N/A	J_{ctrl}^{risk} J_{time}^{risk}	35	45	200	-2.00	-2.00	-6.00

Table 7. Representative control-oriented and runtime-oriented tuned configurations for S3.

Controller	**Optimizer**	$\mathcal{F}(\xi)$	$\mathbf{T_{ini}}$	**N**	$\mathbf{\Delta}$	$\theta_{\mathbf{g}}$	θ_{σ}	$\theta_{\mathbf{thr}}$
C1	MOEA/D	J^{risk}_{ctrl}	16	19	402	-0.63	-4.12	-5.00
		J^{risk}_{time}	11	19	203	-0.30	-4.12	-4.89
C2		J^{risk}_{ctrl}	10	15	431	-0.63	-4.45	-6.88
		J^{risk}_{time}	10	15	227	-0.63	-4.45	-6.88
C3		J^{risk}_{ctrl}	10	29	281	-1.06	-2.82	-3.61
		J^{risk}_{time}	10	10	281	-0.89	-2.82	-3.39
C4		J^{risk}_{ctrl}	11	21	221	-1.25	-2.14	-3.96
		J^{risk}_{time}	13	16	232	0.93	-0.97	-4.66
Ours		J^{risk}_{ctrl}	13	11	529	-1.14	-5.25	-5.91
		J^{risk}_{time}	10	11	200	-1.14	-5.28	-6.08
C1	NSGA-II	J^{risk}_{ctrl}	14	35	421	-0.79	-5.40	-3.04
		J^{risk}_{time}	10	30	259	-0.97	0.16	-5.54
C2		J^{risk}_{ctrl}	12	20	372	-0.89	-4.05	-2.64
		J^{risk}_{time}	12	15	249	0.84	-3.47	-6.96
C3		J^{risk}_{ctrl}	15	17	545	-0.64	-4.02	-3.37
		J^{risk}_{time}	13	11	311	0.20	-4.69	-6.05
C4		J^{risk}_{ctrl}	12	27	302	-1.23	-3.33	-5.72
		J^{risk}_{time}	12	18	245	0.11	-1.61	-6.04
Ours		J^{risk}_{ctrl}	12	13	447	-1.01	-3.17	-3.85
		J^{risk}_{time}	10	14	206	-0.74	-0.98	-2.00
Fixed	N/A	J^{risk}_{ctrl}, J^{risk}_{time}	35	45	200	-2.00	-2.00	-6.00

Table 8. Representative control-oriented and runtime-oriented tuned configurations for S4.

Controller	**Optimizer**	$\mathcal{F}(\xi)$	$\mathbf{T_{ini}}$	**N**	$\mathbf{\Delta}$	$\theta_{\mathbf{g}}$	θ_{σ}	$\theta_{\mathbf{thr}}$
C1	MOEA/D	J^{risk}_{ctrl}	29	17	205	-1.05	-1.60	-3.00
		J^{risk}_{time}	41	25	501	-3.75	-2.10	-1.95
C2		J^{risk}_{ctrl}	21	43	315	-1.39	-4.25	-6.86
		J^{risk}_{time}	10	21	261	-1.39	-4.25	-6.86
C3		J^{risk}_{ctrl}	15	12	291	-2.40	-1.80	-6.72
		J^{risk}_{time}	11	12	291	-2.40	-1.80	-6.72
C4		J^{risk}_{ctrl}	20	35	478	0.34	-1.40	-4.86
		J^{risk}_{time}	26	45	203	0.83	2.00	-4.84
Ours		J^{risk}_{ctrl}	30	26	575	-1.82	-5.60	-4.55
		J^{risk}_{time}	30	26	575	-1.82	-5.60	-4.55
C1	NSGA-II	J^{risk}_{ctrl}	27	39	243	-1.22	0.96	-5.75
		J^{risk}_{time}	10	24	227	0.57	0.61	-3.38
C2		J^{risk}_{ctrl}	12	33	379	-1.54	1.24	-2.85
		J^{risk}_{time}	12	13	226	-0.50	-3.34	-4.04
C3		J^{risk}_{ctrl}	20	19	291	-1.35	-1.95	-3.82
		J^{risk}_{time}	20	10	372	-2.29	0.61	-4.69
C4		J^{risk}_{ctrl}	13	43	397	-1.12	-5.33	-6.90
		J^{risk}_{time}	15	14	225	-0.04	-5.33	-4.67
Ours		J^{risk}_{ctrl}	31	38	444	-2.43	-1.70	-6.12
		J^{risk}_{time}	14	12	239	-0.60	0.22	-3.70
Fixed	N/A	J^{risk}_{ctrl}, J^{risk}_{time}	35	45	200	-2.00	-2.00	-6.00

Table 9. Summary of dominant perturbation directions on S1 for control loss, runtime, retained order, and safety headroom.

Optimizer	$\mathcal{F}(\xi)$	ξ	max $\lvert\delta J_{ctrl}^{risk}\rvert$ ($\times 10^6$)	max $\lvert\delta J_{time}^{risk}\rvert$ (ms)	max $\lvert\delta r_a\rvert$	max $\lvert\delta h_{\min}\rvert$
MOEA/D	J_{ctrl}^{risk}	T_{ini}	4.592	9.332	2	0.077
		N	0.005	10.167	2	0.000
		σ_{thr}	0.036	9.693	2	0.001
	J_{time}^{risk}	T_{ini}	4.596	2.211	2	0.078
		σ_{thr}	0.229	1.219	2	0.005
NSGA-II	J_{ctrl}^{risk}	T_{ini}	4.248	0.949	2	0.090
		N	4.250	0.903	2	0.084
		λ_σ	6.101	2.022	0	0.119
		σ_{thr}	0.257	1.028	1	0.006
	J_{time}^{risk}	T_{ini}	3.335	0.806	2	0.074

Table 10. Summary of dominant perturbation directions on S2 for control loss, runtime, retained order, and safety headroom.

Optimizer	$\mathcal{F}(\xi)$	ξ	max $\lvert\delta J_{ctrl}^{risk}\rvert$ ($\times 10^6$)	max $\lvert\delta J_{time}^{risk}\rvert$ (ms)	max $\lvert\delta r_a\rvert$	max $\lvert\delta h_{\min}\rvert$
MOEA/D	J_{ctrl}^{risk}	T_{ini}	0.921	4.043	4	0.051
		N	0.007	9.636	4	0.000
		$\log_{10}\lambda_g$	0.745	0.825	0	0.041
	J_{time}^{risk}	$\log_{10}\sigma_{thr}$	1.767	2.827	2	0.145
		T_{ini}	0.775	2.110	4	0.043
		N	0.033	0.846	4	0.002
NSGA-II	J_{ctrl}^{risk}	T_{ini}	1.073	2.616	4	0.059
		$\log_{10}\lambda_g$	0.716	1.013	0	0.038
		N	0.036	1.449	4	0.002
	J_{time}^{risk}	T_{ini}	1.073	0.814	4	0.059
		$\log_{10}\lambda_g$	0.716	1.326	0	0.038
		$\log_{10}\sigma_{thr}$	0.399	1.259	2	0.022

Table 11. Summary of dominant perturbation directions on S3 for control loss, run-time, retained order, and safety headroom.

Optimizer	$\mathcal{F}(\xi)$	ξ	**max** $\lvert\delta J^{risk}_{ctrl}\rvert$ ($\times 10^6$)	**max** $\lvert\delta J^{risk}_{time}\rvert$ **(ms)**	**max** $\lvert\delta r_a\rvert$	**max** $\lvert\delta h_{\min}\rvert$
MOEA/D	J^{risk}_{ctrl}	$\log_{10}\lambda_g$	64889.788	46.501	0	0.016
		T_{ini}	0.031	3.329	4	0.000
		N	0.013	1.522	4	0.000
	J^{risk}_{time}	T_{ini}	0.018	1.028	4	0.000
		$\log_{10}\sigma_{thr}$	0.012	0.429	1	0.000
		Δ	0.009	0.648	0	0.000
NSGA-II	J^{risk}_{ctrl}	T_{ini}	16726.286	31.068	4	0.011
		N	16503.198	14.799	4	0.011
		$\log_{10}\lambda_g$	37549.364	13.865	0	0.011
	J^{risk}_{time}	$\log_{10}\sigma_{thr}$	0.069	8.668	1	0.001
		$\log_{10}\lambda_g$	0.128	1.187	0	0.001
		N	0.008	5.608	4	0.000

Table 12. Summary of dominant perturbation directions on S4 for control loss, run-time, retained order, and safety headroom.

Optimizer	$\mathcal{F}(\xi)$	ξ	**max** $\lvert\delta J^{risk}_{ctrl}\rvert$ ($\times 10^6$)	**max** $\lvert\delta J^{risk}_{time}\rvert$ **(ms)**	**max** $\lvert\delta r_a\rvert$	**max** $\lvert\delta h_{\min}\rvert$
MOEA/D	J^{risk}_{ctrl}	T_{ini}	2.804	1.480	2.000	0.016
		N	0.110	4.766	2.000	6.449×10^{-4}
		$\log_{10}\lambda_\sigma$	1.540	2.268	0.000	0.010
	J^{risk}_{time}	T_{ini}	2.804	1.243	2.000	0.016
		$\log_{10}\lambda_\sigma$	1.540	1.380	0.000	0.010
		N	0.110	1.403	2.000	6.449×10^{-4}
NSGA-II	J^{risk}_{ctrl}	T_{ini}	2.400	4.048	2.000	0.015
		N	0.038	5.577	2.000	2.151×10^{-4}
		$\log_{10}\lambda_\sigma$	1.420	3.362	0.000	0.008
	J^{risk}_{time}	T_{ini}	2.824	0.319	2.000	0.016
		N	6.100×10^{-7}	1.540	2.000	3.456×10^{-9}
		$\log_{10}\lambda_g$	6.005×10^{-5}	2.330	0.000	3.402×10^{-7}

Algorithm 2: Risk-aware evolutionary tuning of reduced-order DeePC

Input: Search domain Ξ; population size P; generation budget G; optimizer type $\mathcal{A} \in \{\texttt{MOEA/D}, \texttt{NSGA-II}\}$; MOEA/D parameters T_n, F_{DE}, CR, $\{\lambda^{(i)}\}_{i=1}^{P}$, (s_1, s_2); NSGA-II parameters p_c, p_m, η_c, η_m, p_{reset}, η_{reg}.

Output: Approximate feasible Pareto archive $\mathcal{P}$; generation-wise history $\mathcal{H}$; rollout diagnostics.

Initialize the parent population $\mathcal{P}^{(0)} = \{\xi_i^{(0)}\}_{i=1}^{P} \subset \Xi$.
for $i = 1, \ldots, P$ **do**
    Evaluate $\xi_i^{(0)}$ using Algorithm 1 and store $\mathcal{F}(\xi_i^{(0)})$, $CV(\xi_i^{(0)})$, and the signed-margin diagnostics.
end for
Initialize the feasible nondominated archive $\mathcal{P} \leftarrow ND(\{\xi_i^{(0)} : CV(\xi_i^{(0)}) = 0\})$ and record the initial history statistics.
if $\mathcal{A} = \texttt{MOEA/D}$ **then**
    Construct the weight-space neighborhoods $B(i)$ using Eq. (15).
end if
for $g = 0, \ldots, G-1$ **do**
    if $\mathcal{A} = \texttt{MOEA/D}$ **then**
        Update the ideal point $z^{\star,(g)}$.
        for $i = 1, \ldots, P$ **do**
            Generate the continuous and structural trail-vector parts using Eqs. (17) and (18).
            Perform crossover and mixed-variable repair onto Ξ.
            Evaluate the offspring $\tilde{\xi}_i^{(g)}$ using Algorithm 1.
            foreach $j \in B(i)$ **do**
                Replace incumbents using the feasibility-first rule in Eq. (19) and the normalized Tchebycheff scalarization in Eq. (16).
            end foreach
        end for
        Set the next parent population to the neighborhood-updated population.
    else if $\mathcal{A} = \texttt{NSGA-II}$ **then**
        Select parents using the tournament rule in Eq. (24).
        Generate discrete and continuous offspring using Eqs. (25)–(29).
        Evaluate all offspring using Algorithm 1.
        Perform constrained nondominated sorting using Eq. (20) and augmented crowding using Eq. (23).
        Set the next parent population to the selected elite population.
    Update the feasible archive $\mathcal{P} \leftarrow ND\big(\mathcal{P} \cup \{\xi_i^{(g+1)} : CV(\xi_i^{(g+1)}) = 0\}\big)$.
    Record generation-wise history statistics $\mathcal{H}^{(g+1)}$.
end for
return *$\mathcal{P}$, $\mathcal{H}$, and the rollout diagnostics.*

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